Name: Amritansh Srivastava
Reg No: 2427030667

INDEX

S.No.	Name of the Program	Date	Submission Date	Sign.
1	Identifying the Requirements from Problem Statement.			
2	Requirement Analysis and development of Software Requirement Specification (SRS) Sheet for suggested system.			
3	Estimation of Project Metrics (Effort Estimation).			
4	Estimation of Project Metrics (Cost Estimation).			
5	Modelling UML Use Case Diagram and Class Diagram for the suggested system.			

Experiment - 1

AIM: Identifying the Requirements from the Problem Statement.

REQUIREMENTS:

Hardware Interfaces

- Processor: Intel i5 (or equivalent) and above
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 256 GB SSD or higher
- Internet connectivity for cloud-based backend (Supabase) access

Software Interfaces

- Any modern operating system (Windows / macOS / Linux)
- Code editor - VS Code
- Web Browser (Chrome / Edge / Firefox) with internet access

THEORY:

Title: SDC Portal - A Full-Fledged Website with Role-Based Team Management System for the Software Development Centre, Manipal University Jaipur

Problem Statement: The Software Development Centre (SDC) at Manipal University Jaipur is a student-driven body that works on real-world software projects, hackathons, and technical initiatives. Currently, there is no unified digital platform that represents the SDC, showcases its work, and manages its internal team operations. Team coordination - assigning tasks and projects to interns, tracking their progress, and managing members across three levels of responsibility (Intern, Admin and Super Admin) - is presently handled through informal, non-centralized means, which makes tracking accountability and productivity difficult.

Identification of Requirements:

- Centralized Platform: A single web platform that serves as the public face of SDC (projects, events, achievements) and hosts an internal Team Portal for operations.
- Role-Based Access Control (RBAC): Three access levels - Intern, Admin and Super Admin - each with a distinct, clearly separated set of permissions.
- Task & Project Assignment: Super Admins and Admins must be able to create, assign and update tasks/projects for interns.
- Progress Monitoring: Admins and Super Admins should be able to monitor the real-time status of tasks assigned to interns.
- Self-Service for Interns: Interns should be able to view assigned tasks and update their own task status.
- Secure Authentication: Login, logout, session management and password management for all roles, handled through Supabase Auth.
- Real-Time Data: The database must reflect task/project updates in real time using Supabase's real-time subscriptions.
- File & Document Storage: Ability to store and retrieve project-related documents and images via Supabase Storage.
- Responsiveness: The public website and portal must be responsive across desktop and mobile devices.
- Scalability & Maintainability: The system must be built with a component-based, modular architecture (React.js) so that new features can be added with minimal rework.

Experiment - 2

AIM: Do requirement analysis and develop the Software Requirement Specification (SRS) Sheet for the SDC Portal.

REQUIREMENTS:

Hardware Requirements:

- Processor: Intel i5 (or equivalent) and above
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 256 GB SSD or higher
- Internet connectivity for cloud-based backend (Supabase) access

Software Requirements:

Front-End:

- React.js - Component-based library for building the user interface
- Tailwind CSS - Utility-first CSS framework for responsive and modern styling
- React Router - For client-side routing and navigation
- Vite - Build tool and development server
- Recharts / Chart.js - For visualising task and progress statistics on dashboards

Back-End:

- Supabase - Backend-as-a-Service providing PostgreSQL database, Authentication, Row Level Security (RLS) for role-based access, and real-time subscriptions
- Supabase Auth - For secure login and session management of Intern, Admin, and Super Admin roles
- PostgreSQL (via Supabase) - Relational database for storing user, role, task, and project data
- Supabase Storage - For storing documents, images, and other project-related files

Other Tools:

- VS Code - Code editor
- Git & GitHub - Version control and collaboration
- Vercel / Netlify - Deployment and hosting of the front-end application
- Figma - UI/UX design and prototyping

THEORY:

Functional Requirements:

- The system shall allow users to log in and log out securely based on their assigned role.
- The system shall allow Super Admins to create Admin and Intern accounts and assign roles.
- The system shall allow Admins/Super Admins to create projects and assign tasks to interns.
- The system shall allow interns to view their assigned tasks and update task status.
- The system shall display real-time dashboards showing task and project progress.
- The system shall allow storage and retrieval of project-related files/documents.
- The system shall display public information about SDC (projects, events, achievements) to visitors.

Non-Functional Requirements:

- Security: Role-based access must be enforced using Row Level Security (RLS) at the database level.

- Performance: Dashboard and task updates should reflect within a few seconds using real-time subscriptions.
- Usability: The interface must be intuitive and responsive across screen sizes.
- Availability: The cloud-hosted application should be accessible 24x7 (barring host downtime).
- Maintainability: Component-based front-end architecture to keep the codebase modular and easy to extend.
- Scalability: Backend-as-a-Service (Supabase) allows the system to scale as the number of users/interns grows.

Experiment - 3

AIM: Estimation of Project Metrics (Effort Estimation) for the SDC Portal.

Effort Estimation using COCOMO Model (Organic Mode)

The SDC Portal is a small-to-medium, well-understood web application developed by a small, experienced team, and is therefore classified as an Organic project under the Basic COCOMO model.

Estimated size of the project: KLOC = 12 (thousand lines of code)

For an Organic mode project, the constants used are:

- a = 2.4
- b = 1.05

The Basic COCOMO effort equation is:

E = a × (KLOC)^b person-months

Substituting the values:

E = 2.4 × (12)^1.05

E = 2.4 × 13.585

E ≈ 32.6 person-months

So, the estimated effort required for the SDC Portal is approximately 32.6 person-months.

Development Time Estimation

The Basic COCOMO development-time equation for Organic mode is:

Tdev = 2.5 × (E)^0.38

Substituting E = 32.6:

Tdev = 2.5 × (32.6)^0.38

Tdev = 2.5 × 3.76

Tdev ≈ 9.4 months

Average number of persons required = E / Tdev = 32.6 / 9.4 ≈ 3 to 4 persons.

Experiment - 4

AIM: Estimation of Project Metrics (Cost Estimation) for the SDC Portal.

COST ESTIMATION

Average Productivity is assumed to be 500 LOC / PM (Lines of Code per Month).

Average Labor Cost is assumed to be Rs. 15,000 per month (typical for a student/entry-level developer stipend rate used for academic estimation).

Total Estimated Project Cost can be calculated as follows:

Cost = Effort × Labor Rate

Considering the Effort estimated in Experiment - 3, E ≈ 32.6 person-months:

Cost = 32.6 × Rs. 15,000

Cost ≈ Rs. 4,89,000

So, the estimated cost of developing the SDC Portal is approximately Rs. 4,89,000.

This estimation provides a rough idea of the total project cost. Actual costs may vary based on factors such as hosting/cloud charges (Supabase, Vercel/Netlify), labor rates, additional resources, and unforeseen circumstances during the project lifecycle. Regular monitoring and adjustment of the project budget are necessary to ensure financial viability and successful project delivery.

Experiment - 5

AIM: To prepare the UML Use Case Diagram and Class Diagram for the SDC Portal.

REQUIREMENTS:

Hardware Interfaces

- Intel i5 (or equivalent) processor, 8 GB RAM or higher
- Screen resolution of at least 1366 x 768 for proper viewing of diagrams

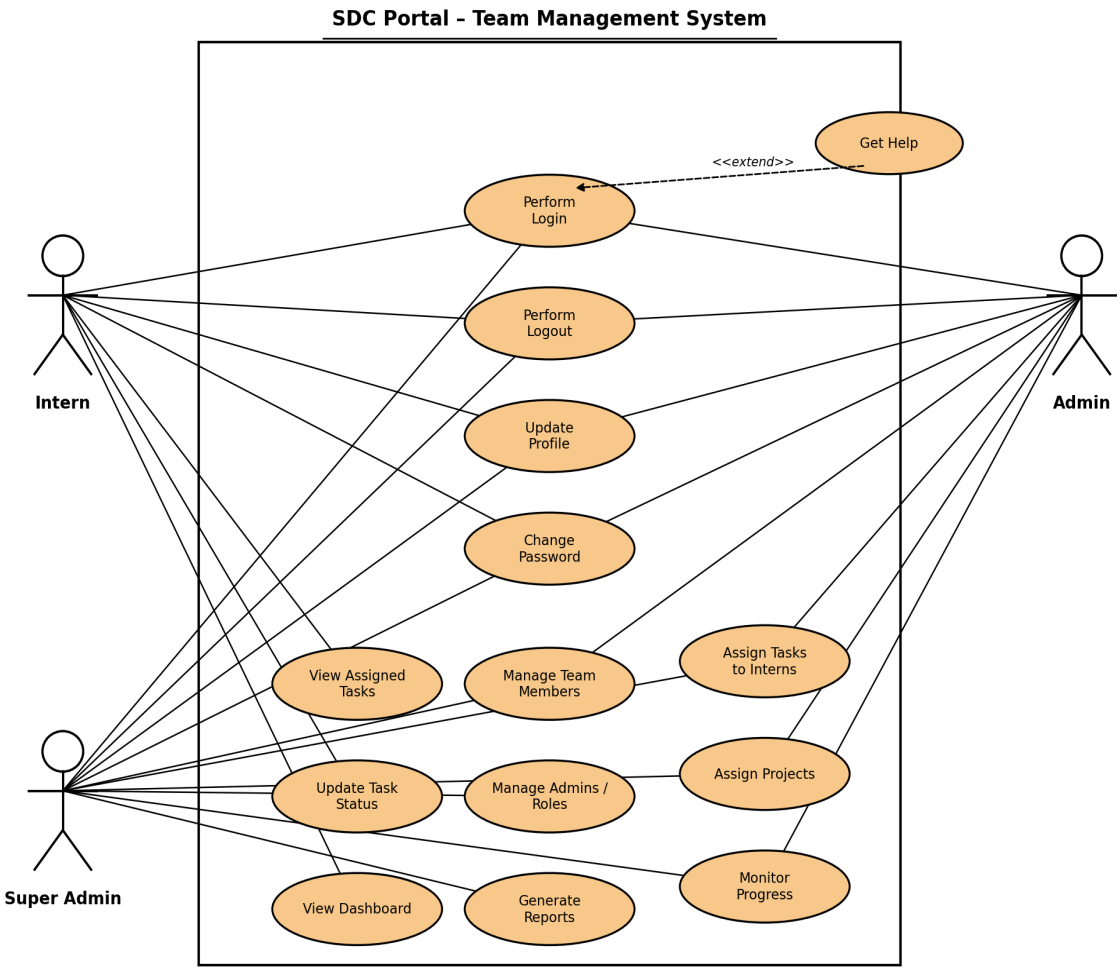
Software Interfaces

- Any modern operating system (Windows / macOS / Linux)
- UML diagramming tool (e.g., Draw.io, Lucidchart, StarUML) or equivalent

THEORY

Use-Case Diagram for SDC Portal:

A use-case diagram captures the functional requirements of a system by showing the interactions between actors and the system's use cases. For the SDC Portal, three actors were identified based on the Role-Based Access Control (RBAC) design: Intern, Admin and Super Admin. Common use cases such as Login, Logout, Update Profile and Change Password are shared across all roles, while Admins and Super Admins additionally handle task/project assignment and progress monitoring, and Super Admins alone manage other Admins and roles.

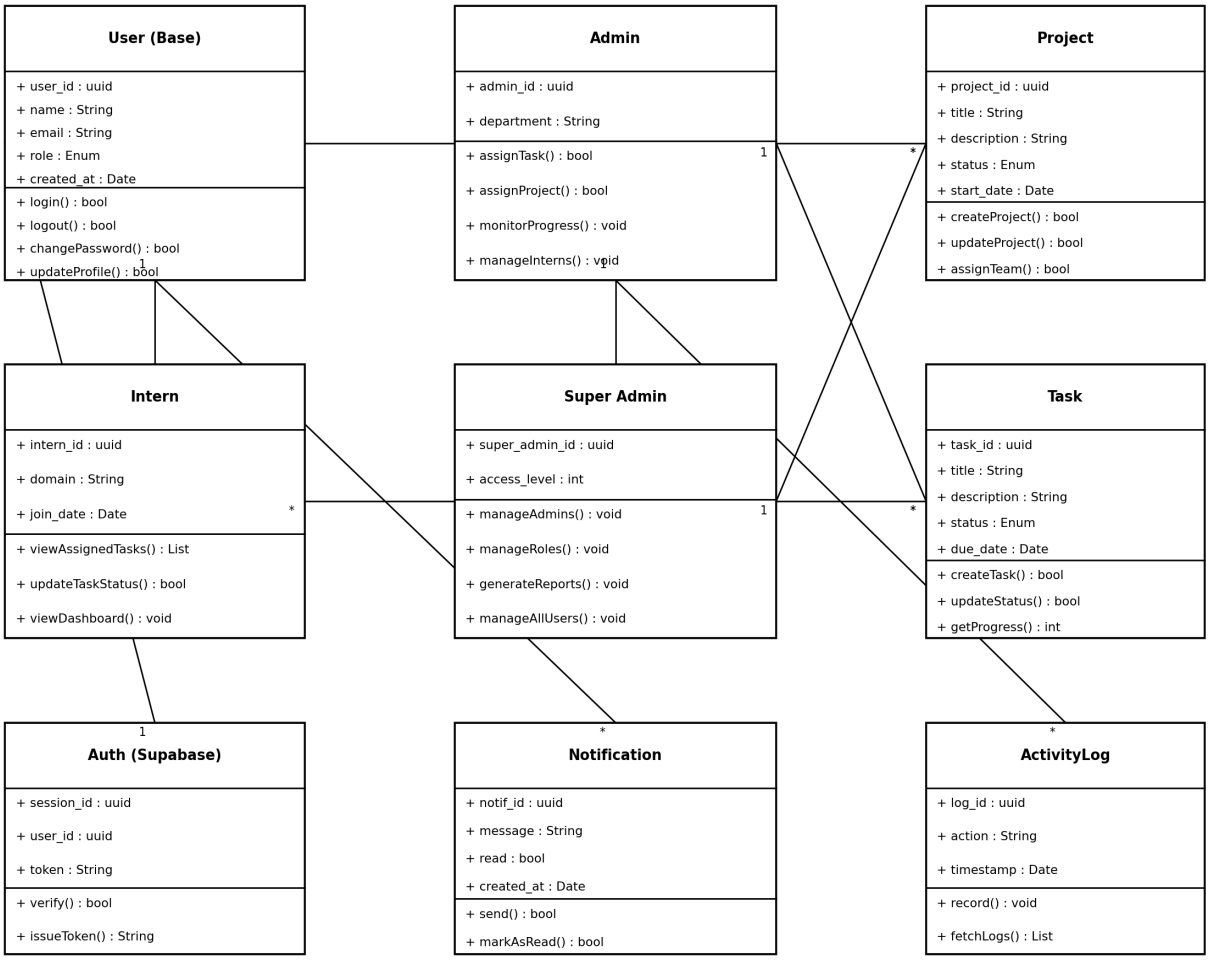


Use-Case Diagram for SDC Portal

Class Diagram for SDC Portal:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects. It is the main building block of object-oriented modelling and is used for both conceptual modelling and for translating the model into programming code. The class diagram for the SDC Portal is shown below - the base User class is extended by Intern and Admin, and Super Admin further extends Admin; Admins and Super Admins are associated with Project and Task classes, while User is associated with Auth, Notification and ActivityLog classes.

CLASS DIAGRAM FOR SDC PORTAL – TEAM MANAGEMENT SYSTEM



Class Diagram for SDC Portal